



ENGIN 604 INTRODUCTION TO PYTHON FOR FINANCE — FALL 2021

HOMEWORK 3

Due: 11:59pm, Saturday, April 10

Submit to: engin604assignments@gmail.com

Maximum page limit: 5 pages

See the homework policies at <https://docenciaweb.fen.uchile.cl>



1. Import the files `stocks.pkl`, `sp500.pkl` and `rfree.pkl`.
 - `stocks.pkl`: Facebook, Apple, Amazon, Netflix and Google indices from 2014:12 to 2020:12.
 - `sp500.pkl`: Closing price of the S&P 500 index from 2014:12 to 2020:12.
 - `rfree.pkl`: Risk-free rate from 2015:01 to 2020:12.
2. Repeat the exercise from the **Application: Capital Asset Pricing Model** guide (up to question 6) using logarithmic returns instead of arithmetic returns.

$$R_{i,t} = \ln \left(\frac{P_{i,t}}{P_{i,t-1}} \right) \quad i = 1, \dots, 5 \quad t = 1, \dots, 73$$

Hint: NumPy has the `np.log()` function to compute the natural logarithm.

3. Show that the CAPM $\hat{\beta}$ is equal to:

$$\hat{\beta} = \sum_{i=1}^{n=5} w_i * \hat{\beta}_i$$

Where w_i is the weight of asset i in the portfolio created in (2) and $\hat{\beta}_i$ the second estimated coefficient of the following simple linear regression:

$$R_{i,t} - R_f = \alpha_i + \beta_i(R_{m,t} - R_f) + \varepsilon_{i,t}$$

Where $R_{i,t}$ is the logarithmic return of asset i .

Hint: Remember that the guide contains the function to obtain $\hat{\beta}$ from its matrix form.

4. Considering only the logarithmic return (without subtracting R_f) of the assets in (2):
 - a. Store the average return of each one in an *array*.
 - b. Compute the variance-covariance matrix.
5. The global minimum variance portfolio is the one with the lowest possible risk for a given level of return; in other words, there is no other portfolio for that given level of return that can have a lower risk. The optimization problem expressed in its matrix form is:

$$\min_{\mathbf{m}} \sigma_{p,m}^2 = \mathbf{m}'\Sigma\mathbf{m} \quad s.t \quad \mathbf{m}'\mathbf{1} = 1$$

Whose solution (also in matrix form) is:

$$\mathbf{m} = \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}'\Sigma^{-1}\mathbf{1}}$$

Where $\mathbf{m}$ is the vector of optimal weights of the minimum variance portfolio, $\mathbf{1}$ a vector of ones whose dimension is $n \times 1$, with n being the total number of assets, and Σ^{-1} the inverse of the variance-covariance matrix ($n \times n$).

Using the matrix solution for $\mathbf{m}$, write a function whose *inputs* are: the vector of average returns and the variance-covariance matrix of the assets that would make up the minimum variance portfolio.

The *output* of the function must be the weights $\mathbf{m}$, the expected return and the standard deviation of the minimum variance portfolio.

- The expected return of the minimum variance portfolio is:

$$R_p = \mathbf{m}' \cdot \mu$$

Where μ is the vector of average returns of dimension $n \times 1$.

- The standard deviation of the minimum variance portfolio is:

$$\sigma_{p,m} = \sqrt{\mathbf{m}'\Sigma\mathbf{m}}$$

Test the function using the vector of returns created in (4.a) and the variance-covariance matrix created in (4.b). Explain your function line by line.